

KMP Workshop: Setup Before the Session

Sep 29, 2026 · @Tomer

Why set up ahead

Please finish this setup before the session on [session date]. The first time you open the project, Gradle downloads itself, parts of the Android SDK and every library the project uses. That can take 10–20 minutes, or longer on a slow or proxied connection.

If everyone arrives with the project already synced and running, we spend the two hours writing Kotlin instead of watching progress bars. Plan for about 45 minutes, most of it waiting.

Requirements

Any Windows, macOS or Linux laptop works for the workshop. iOS is optional and needs a Mac.

What	Requirement	Notes
Memory	16 GB RAM recommended	8 GB works, but the emulator will be slow
Disk space	About 15 GB free	For Android Studio, the SDK, an emulator image and Gradle caches
JDK	17 or newer	The JDK bundled with Android Studio is fine
IDE	Android Studio, latest stable	IntelliJ IDEA also works
IDE plugin	Kotlin Multiplatform	Installed from the plugin Marketplace
Android device	An emulator, or a phone with USB debugging on	Either is fine
Git	Any recent version	To clone the workshop repo
Network	Access to the sites listed below	Check this if you're on the office network or VPN

What	Requirement	Notes
iOS (optional)	A Mac with the latest Xcode	Only if you want to run the iOS app

Gradle needs to reach these addresses during the sync:

- services.gradle.org (downloads Gradle itself)
- repo.maven.apache.org (Maven Central)
- dl.google.com and maven.google.com (Google Maven and the Android SDK)
- plugins.gradle.org (the Gradle Plugin Portal)

The final lab also calls the free weather API at api.open-meteo.com.

Setup, step by step

Steps 5 and 6 are the ones that matter most: they do the long downloads, so please don't skip them.

1. **Install Android Studio.** Download the latest stable version from developer.android.com/studio. Run the setup wizard with the Standard settings; it installs the Android SDK.
2. **Install the Kotlin Multiplatform plugin.** Open Settings (on macOS: Android Studio › Settings), go to Plugins › Marketplace, search for "Kotlin Multiplatform", install it and restart.
3. **Create an Android emulator.** Open Device Manager, create a device (any recent Pixel is fine) and download its system image now. It's 1-2 GB. Skip this if you'll use a phone with USB debugging.
4. **Clone the workshop repo.** In a terminal, run `git clone https://github.com/TomerGlick/KMP-Starter-Project.git`. You can also use File › New › Project from Version Control in Android Studio.
5. **Open the project and let Gradle sync finish.** Use File › Open and pick the cloned folder (the one containing `settings.gradle.kts`). Choose "Trust Project" if asked.
 - Wait until the sync finishes, even if it takes 20 minutes. Don't cancel it.
 - If Android Studio offers to upgrade the Android Gradle Plugin or Gradle, decline. The versions are pinned to work together.
 - On Windows and Linux you'll see warnings that iOS targets are disabled. That's expected.

6. **Run the desktop app.** In Android Studio's Terminal, run `./gradlew :composeApp:run` (on Windows: `gradlew.bat :composeApp:run`). A window titled "KMP Weather" should open showing "Hello, KMP!". The first run downloads a few more things.
7. **Run the Android app.** Pick the `composeApp` run configuration and your emulator or phone in the toolbar, then click Run. You should see the same "Hello, KMP!" screen.
8. **On the day, pull the latest version.** Run `git pull` before the session starts, in case anything changed.

Am I ready?

You're ready when you can tick all of these:

- ☐ Android Studio and the Kotlin Multiplatform plugin are installed
- ☐ The project is cloned and opened, and Gradle sync finished without errors
- ☐ The desktop app opens and shows "Hello, KMP!"
- ☐ The Android app runs on an emulator or phone and shows "Hello, KMP!"

If any box stays unticked, check Troubleshooting below or contact [your name] before the session, not during it.

Troubleshooting

Most sync failures on a work laptop come from the network: a proxy, the VPN, or SSL inspection.

Problem	What to try
Sync fails with "Could not resolve", "Connection timed out" or "Could not GET"	Try another network (home Wi-Fi or a phone hotspot), or disconnect the VPN. If you're behind a proxy, set it under Settings › HTTP Proxy and accept when Android Studio offers to apply it to Gradle.
Sync fails with "PKIX path building failed"	Your network inspects SSL traffic. Ask IT for the company certificate and how to add it to the JDK that Gradle uses, or sync on another network.

Problem	What to try
"Unsupported class file major version" or "requires Java 17"	Go to Settings › Build, Execution, Deployment › Build Tools › Gradle and set Gradle JDK to 17 or newer (the bundled JetBrains Runtime is fine).
"SDK location not found"	Open the project in Android Studio once so it writes <code>local.properties</code> , or set the <code>ANDROID_HOME</code> environment variable.
<code>./gradlew: Permission denied</code> (macOS or Linux)	Run <code>chmod +x gradlew</code> in the project folder.
Sync seems stuck	It's usually still downloading: check the Build window. If nothing moves for 10 minutes, run <code>./gradlew --stop</code> and sync again.
The emulator won't start or is very slow	Turn on hardware virtualization (Intel VT-x or AMD-V) in the BIOS. On Windows, also enable Windows Hypervisor Platform. Or use a phone with USB debugging.

Still stuck? Send [your name] a screenshot of the error from the Build window.

Optional: iOS

You don't need iOS for the workshop; I'll demo it live. If you have a Mac and want to try it yourself:

1. Install the latest Xcode from the Mac App Store, open it once, accept the license and let it install its components.
2. Follow the "iOS" section of the repo's README. It explains how to add the `iosApp` Xcode project, which isn't included in the repo.
3. Expect the first iOS build to take several minutes.

Questions

Contact Tomer Glick at tomer.glick@bcodes.co.il before the session.